Stimuli and Performance Exploring

1. **Are there levels where performance is extreme** (almost always correct or almost always wrong) **and several intermediate levels where performance is not at ceiling or floor?**

Gap	Total trials	Hits	Hit Rate	D-prime
1	42	10	0.2381	0.1993
2	36	18	0.5000	0.9117
3	42	26	0.6190	1.2147
4	36	21	0.5833	1.1222
5	42	33	0.7857	1.7034
6	42	39	0.9286	2.3770
7	36	34	0.9444	2.5050
8	36	31	0.8611	1.9971
9	36	35	0.9722	2.8262
10	48	45	0.9375	2.4459
11	36	35	1.0000	3.1122

Table 1: Table 1: Hit Rates and Sensitivity Index (d') Across Stimulus Gap Levels

Based on the data in **Table 1**, the performance across the stimulus groups clearly demonstrates both extreme and intermediate levels of sensitivity. The most difficult stimulus, **Gap 1**, shows a near-floor performance with a hit rate of approximately 24% and a low D-prime of 0.1993. Conversely, the easiest stimulus, **Gap 11**, reaches the ceiling with a perfect hit rate of 100% and a high D-prime of 3.1122.

2. **Does hit rate increase as the difference between figures increases?**

Yes, there is a positive trend that has been observed in a correlation graph (*Figure 1*) between the difficulty levels and the *d'*. It gives an indication which stimuli contribute to the variance better.

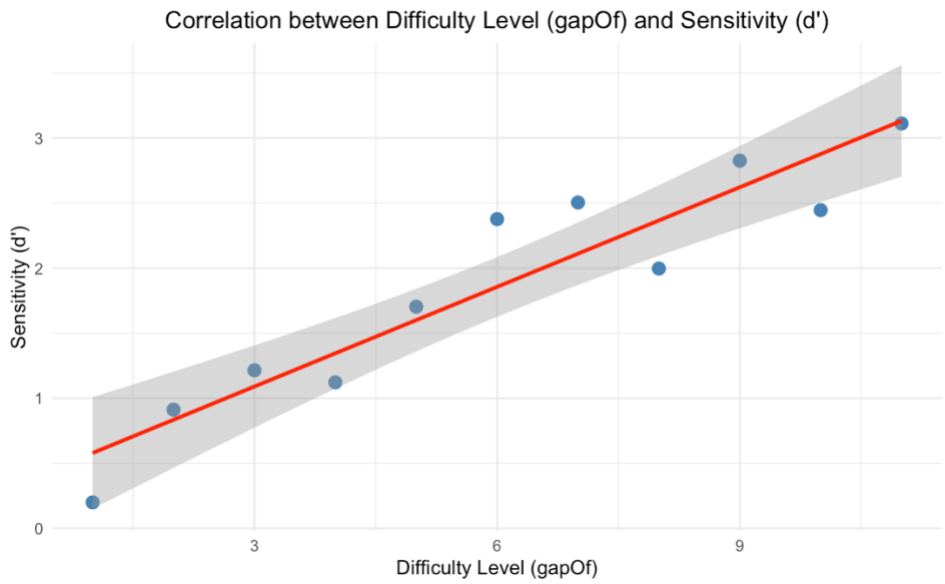

Figure 1: Correlation between difficulty level and d prime.

3. Is there any extreme effect of the orientation?

Conditions	Hit	Miss	% Hit	% Miss
0_vs_minus30	45	15	75	25
0_vs_plus30	68	16	80.95	19.04
minus30_vs_0	72	30	70.58	29.41
minus30_vs_plus30	45	15	75	25
plus30_vs_0	41	13	75.92	24.07
plus30_vs_minus30	57	15	79.16	20.83

Table 2: Comparison of Detection Accuracy and Error Rates Across Orientation Conditions

Based on the data in **Table 2**, there is no evidence of an extreme effect related to the orientation conditions. Specifically, the **minus30_vs_plus30** and **plus30_vs_minus30** groups yielded hit rates of 75% and 79.16%, respectively that is between the range.